Faozia Fariha
Researcher, NLP Lab, CUET, Chattogram, Bangladesh
faoziafarihaaa@gmail.com | +880 1309 724762 | Google Scholar | LinkedIn | GitHub

RESEARCH INTERESTS

Natural Language Processing, Large Language Models, Multimodal AI, Green AI & Sustainable ML, Agentic AI Systems, Federated Learning, AI Safety & Security

EDUCATION

Chittagong University of Engineering & Technology (CUET), Chattogram, Bangladesh Jan 2022 – Jun 2026
Bachelor of Science in Computer Science and Engineering Cumulative GPA: 3.84/4.00
Thesis: GreenFed-LoRA: Energy-Aware Federated Fine-Tuning of Large Language Models
Supervisor: Dr. Mohammad Shamsul Arefin, Professor, Department of CSE, CUET
Ranked among the Top **10** students of the batch (out of approx. 120)

ACADEMIC EXPERIENCE

ELITE Research Lab United States (Remote)
Research Intern May 2026 – Present

- Investigate AI agent security with a focus on secure cross-session memory protection for LLM-based agents.
- Design threat models and mitigation strategies for persistent memory vulnerabilities in multi-agent pipelines.

CUET Undergraduate Thesis Project Chattogram, Bangladesh
Principal Researcher Jan 2026 – Jun 2026

- Designed GreenFed-LoRA, a federated fine-tuning framework for LLMs that jointly optimises model utility and environmental sustainability by treating energy consumption and carbon emissions as core optimisation objectives.
- Implemented LoRA-based parameter-efficient fine-tuning within a federated learning architecture, reducing communication overhead and computational cost across heterogeneous clients.
- Evaluated framework performance on benchmark NLP tasks, comparing accuracy–energy trade-offs against centralised and standard federated fine-tuning baselines.

NLP Lab, CUET Chattogram, Bangladesh
Graduate Research Member Nov 2025 – Present

- Collaborate on shared-task submissions for SemEval-2026 (Tasks 1 and 10) and LT-EDI-2025, producing 3 peer-reviewed workshop papers.
- Implement and fine-tune instruction-tuned LLMs (Qwen, LLaMA) for multilingual text generation and psycholinguistic marker extraction.
- Develop multimodal pipelines combining BERT-based text encoders and Vision Transformers for misogyny detection, achieving **6th place worldwide** in LT-EDI 2025 Shared Task.
- Contribute to dataset curation, experimental design, and quantitative evaluation across multiple NLP tasks.

PROJECTS

Multimodal Misogyny Detection using BERT and Vision Transformers Chattogram, Bangladesh
Research Project Jan 2025 – May 2025

- Developed a multimodal hate speech detection system fusing BERT-based textual features with Vision Transformer image embeddings using late-fusion classification.
- Trained and evaluated models on the LT-EDI 2025 shared-task dataset, achieving **6th place worldwide** among all participating systems.
- Published results in the proceedings of the 5th Language, Data and Knowledge conference (LT-EDI Workshop, 2025).

Bengali Physics MCQ Question Answering System Chattogram, Bangladesh
Independent Research Project Jan 2025 – Dec 2025

- Constructed an LLM-based reasoning pipeline for over 1,500 Bengali physics multiple-choice questions using Qwen3-14B via the Hugging Face API.
- Engineered chain-of-thought prompts, achieving **82%** zero-shot accuracy without any fine-tuning or task-specific training data.

CUET Thesis & Project Portal Chattogram, Bangladesh
Lead Developer Jan 2024 – Dec 2024

- Built a centralised thesis and project management portal using React.js and Firebase with authentication and role-based access control for faculty and students.
- Deployed real-time updates and notification systems, serving 200+ active student users.

PUBLICATIONS

Conference & Workshop Papers

Published

1. **F. Fariha**, L. Khan, M. A. Chowdhury, K. Ahmed and M. M. Hoque. "CUET_320 at SemEval-2026 Task 10: Few-Shot Large Language Models for Psycholinguistic Marker Extraction and Conspiracy Detection." In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval 2026)*, 2026.
2. M. A. Chowdhury, L. Khan, **F. Fariha**, S. H. Shohan and M. M. Hoque. "CUET_Clashing at SemEval-2026 Task 1: Multilingual Joke Generation Under Lexical and Topical Constraints Using Small Instruction-Tuned LLMs." In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval 2026)*, 2026.
3. M. Chowdhury, S. Sultana, **F. Fariha**, A. Mallik, I. Ahammed and M. A. Miah. "Maestro: A Psychological Disorder and Matrimony Assistance Smartphone Application with Personal Development and Therapist Suggestion Features." In *2025 International Conference on Computing and Communication Technologies (ICCCT)*, IEEE, 2025.
4. M. Rahman, **F. Fariha**, N. Tabassum, S. Rahman and H. Murad. "CUET_12033@LT-EDI-2025: Misogyny Detection." In *Proceedings of the 5th Conference on Language, Data and Knowledge (LT-EDI Workshop)*, 2025.

SELECTED COURSES

Undergraduate Core Courses

- Artificial Intelligence
- Machine Learning
- Digital Image Processing
- Data Structures & Algorithms
- Database Management Systems
- Information Security
- Big Data Analytics
- Internet of Things

Online / External Courses

- Supervised Machine Learning: Regression and Classification (Coursera)
- Neural Networks and Deep Learning (Coursera)

AWARDS & HONOURS

6th Place Worldwide — Worldwide
LT-EDI 2025 Shared Task on Multimodal Misogyny Detection (international competition) — 2025

Champion — Bangladesh
NodeScape Hackathon — 2025

Top 7 Finalist — Bangladesh
HackCSB Hackathon (among 116+ participating teams) — 2024

Dean's Award — Chattogram, Bangladesh
Awarded for academic excellence in 8 consecutive semesters, Department of CSE, CUET — 2022–2026

Top 10 Academic Rank — Chattogram, Bangladesh
Ranked among the top 10 students of the B.Sc. CSE batch at CUET — 2022–2026

46th Place, National Girls' Programming Contest (NGPC) — Bangladesh
National-level competitive programming competition — 2023

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, JavaScript, SQL
- **AI / ML:** Deep Learning, Natural Language Processing, Large Language Models (LLMs), Vision Language Models (VLMs), Retrieval-Augmented Generation (RAG), Federated Learning, Prompt Engineering
- **Frameworks & Libraries:** PyTorch, Hugging Face Transformers, LangChain, FastAPI, Express.js, React.js
- **Tools & Platforms:** Git, Docker, Linux, LaTeX, Firebase, Power BI
- **Methodologies:** Experimental design, ablation studies, quantitative evaluation, literature synthesis

PROFESSIONAL SERVICE

- **Competitive Programming Mentor:** Solved 500+ problems on Codeforces (@bounty_hunter12, max rating 1225) and LeetCode; mentored junior students in algorithmic problem-solving.
- **Industrial Training (BRAC IT Services Ltd., Nov 2025):** Completed structured training in AI Engineering, DevOps, Cybersecurity, Data Engineering, Software Quality Assurance, and Web Development.

REFERENCES

Dr. Mohammed Moshiul Hoque
Professor, Dept. of CSE
Director, CUET IT Business Incubator
CUET, Chattogram, Bangladesh
moshiul_240@cuet.ac.bd

Dr. Mohammad Shamsul Arefin
Professor, Dept. of CSE
Thesis Supervisor
CUET, Chattogram, Bangladesh
sarefin@cuet.ac.bd